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Here we present Russell's paradox, which shows that a very natural assumption about sets leads to a contradiction.

The Comprehension Assumption

We begin with the following general assumption:

Comprehension Assumption. For any condition $P(x)$, there exists a set

$$S = \{x \mid P(x)\}$$

such that, for every object a ,

$$a \in S \quad \text{if and only if} \quad P(a) \text{ holds.}$$

Informally: for any property you can describe, there is a set consisting of exactly the objects that have that property.

The Russell Set

Consider the specific property

$$P(x) : \iff x \notin x.$$

Thus $P(x)$ means “ x is a set that does not contain itself as a member.”

By the comprehension assumption, there exists a set consisting of all sets that do not contain themselves. Call this set R :

$$R = \{x \mid x \notin x\}.$$

By the definition of R , for any set a ,

$$a \in R \quad \iff \quad a \notin a.$$

This equivalence is the key to the paradox.

Is R a Member of Itself?

We now ask: is $R \in R$? We will see that both possible answers lead to a contradiction.

Case 1: Assume $R \in R$. Suppose first that $R \in R$. Then, by the defining property of R ,

$$R \in R \iff R \notin R.$$

Thus, from the assumption $R \in R$, we must conclude $R \notin R$. Hence

$$R \in R \Rightarrow R \notin R,$$

which is a contradiction.

Case 2: Assume $R \notin R$. Suppose instead that $R \notin R$. Again, by the definition of R ,

$$R \in R \iff R \notin R.$$

Now we start from $R \notin R$. But the equivalence states that $R \notin R$ holds if and only if $R \in R$. Thus

$$R \notin R \Rightarrow R \in R,$$

which is also a contradiction.

Conclusion

In summary, we have shown:

- If $R \in R$, then $R \notin R$.
- If $R \notin R$, then $R \in R$.

Thus both possibilities for the statement $R \in R$ lead to contradictions. Therefore, assuming that such a set R exists is impossible.

But the existence of R followed directly from the comprehension assumption. Hence we conclude that the unrestricted comprehension assumption cannot be maintained without contradiction. This contradiction is known as *Russell's paradox*.

Abstract

Embrace the Unknown: Russell's Paradox and The Halting Problem

What to do when a very natural assumption about sets leads to a contradiction? A disastrous revelation that threatened foundations of mathematics at the end of the 19th century, Russell's Paradox uncovers our naive assumption about what we can or cannot do with sets. In this talk, we inspect the Paradox and its Theory of Computation counterpart, The Halting Problem using a proof reminiscent to Cantor's diagonalization argument.